

Effective mass from spanning trees, and the absence of Bloch phases on rank-two hyperbolic lattices

Ruqing Chen^{1,*}

¹ GUT Geoservice Inc., Rivière-Beaudette, Québec J0P 1R0, Canada

(Dated: September 18, 2026)

Every $\mathbb{Z}^d$ -periodic tight-binding model with a finite unit cell is an abelian cover of a finite graph, the winding numbers of the hoppings playing the role of a cocycle. We show that for such a lattice the inverse effective-mass tensor at the bottom of the band is, exactly, the Albanese metric of the unit cell: $(M^{-1})_{ab} = (2t/n)\langle h_a, h_b \rangle$, where h_a is the harmonic projection of the a -th winding cochain and n is the number of sites per cell. Three consequences follow. (i) When the flux threads a single link e , the effective mass has a closed combinatorial form, $M^* = n\kappa_0/[2t\tau(Y_0 - e)]$, where κ_0 is the number of spanning trees of the unit cell and $\tau(Y_0 - e)$ the number of those avoiding e . (ii) The tensor $\kappa_0(n/2t)M^{-1}$ has integer entries: *effective masses on such lattices are quantized*, in units fixed by a spanning-tree count. (iii) The diffusion tensor of the classical walk on the same lattice obeys $\sigma^2 = M^{-1}/[t(q+1)]$ with $q+1$ the coordination number — an Einstein relation with no other non-universal factor. We then prove a no-go theorem in the opposite direction. For a lattice whose unit cell is a free quotient of a thick $\tilde{A}_2$ building — the rank-two analogue of the trees and $\{p, q\}$ tessellations used in circuit-QED hyperbolic lattices — the first Betti number vanishes, so no $\mathbb{Z}^d$ cover exists at all: there is no Brillouin zone, no crystal momentum, and no band structure, and by property (T) no continuous family of unitary characters deforms the trivial one either. The threadable fluxes form a *finite* group, computed exactly as $(\mathbb{Z}/2)^6$, $(\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14$ and $(\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$ on complexes with 21, 24 and 42 sites; on the first of these only π -fluxes exist. Hyperbolic band theory does not survive the passage from rank one to rank two. All statements are verified by machine: 90 checks, exact except where labelled.

I. INTRODUCTION

Band theory is a statement about $\mathbb{Z}^d$ symmetry. A tight-binding model on a lattice with a finite unit cell and a $\mathbb{Z}^d$ translation group decomposes over a Brillouin zone $\mathbb{T}^d$, and the low-energy physics is controlled by the curvature of the lowest band at its minimum — the effective mass. Two recent developments make it worth asking how far this picture is combinatorial rather than geometric.

The first is the experimental realization of *hyperbolic lattices* in circuit quantum electrodynamics [1], where a coplanar-waveguide network is laid out as a $\{p, q\}$ tessellation of the hyperbolic plane, and the subsequent construction of a hyperbolic band theory [2, 3] in which the role of the Brillouin zone is played by a higher-dimensional Jacobian and by a character variety of the (non-abelian) translation group. The platform has since broadened to electrical circuits [4], integrated photonics [5], and scalable superconducting architectures with the unit cell on a genus-three surface [6]. The unit cell of such a lattice is a finite graph, and what is threaded through it is flux.

The second is the observation, standard in the mathematics of crystal lattices since Kotani and Sunada [7, 8], that a $\mathbb{Z}^d$ -periodic graph carries a canonical flat metric on H^1 of its unit cell — the Albanese metric — which controls the long-time heat kernel and the central limit theorem of the random walk, and that this metric is bound

up with the *complexity* of the unit cell, i.e. its spanning-tree count [9]; explicit Kirchhoff expressions for it were given by Baker and Faber [10]. What does not seem to have been said is that this metric *is* the inverse effective-mass tensor of the tight-binding band bottom, exactly and without approximation, and that reading it so turns those Kirchhoff expressions into a constraint on a measurable quantity. That is the content of Secs. III–IV, whose sharpest consequence is a quantization statement (Corollary 5): the entries of M^{-1} , measured in the natural unit $2t/(n\kappa_0)$, are integers.

The paper then turns the question around. Trees and $\{p, q\}$ tessellations are rank-one objects: the tree is the Bruhat–Tits building of GL_2 over a local field, and its finite quotients are ordinary finite graphs, which always have $b_1 \geq 2$ and therefore always admit $\mathbb{Z}^d$ covers. The natural rank-two generalization is a two-dimensional building of type $\tilde{A}_2$ — a simplicial complex of sites, links and plaquettes in which every vertex link is the incidence graph of a projective plane — whose finite quotients are the Ramanujan complexes [14]. These are the higher-rank hyperbolic lattices. We show in Sec. VI that for them band theory does not merely become harder, it ceases to exist: $b_1 = 0$ by Garland’s vanishing theorem [12] as extended by Ballmann and Świątkowski [13], so there is no $\mathbb{Z}^d$ cover, no crystal momentum, and no continuous flux parameter. The Aharonov–Bohm phases that a rank-two hyperbolic lattice admits form a finite abelian group, which we compute exactly on three complexes.

The two halves meet at a single graph. The Heawood graph — the incidence graph of the Fano plane

* ruqing@hotmail.com

— is a perfectly good rank-one unit cell: it has $b_1 = 8$, $50421 = 3 \cdot 7^5$ spanning trees, and a single-link flux gives it the effective mass $M^* = 147/(8t)$ (Table I). It is also exactly the vertex link of a thick $\tilde{A}_2$ building of order two. Assembling Heawood links into a rank-two complex destroys b_1 completely (Table III). The obstruction is not local.

Relation to previous work

Theorem 2 lies close to known mathematics. Once the Albanese metric is identified with the covariance of the crystal-lattice central limit theorem [7] and with a Kirchhoff ratio [9, 10], Eq. (4) follows from a second-order perturbation argument, which we give in full for completeness rather than for novelty. What we claim as new is: the tensor form of Theorem 2 together with the explicit statement that this object is the physical effective-mass tensor; the quantization Corollary 5 and its reading as a rigid constraint on measurable band curvature; the deletion–contraction closed form (8); the loop-flux Corollary 6; and the whole of Sec. VI. The closest prior result in the physics literature that we are aware of is that of Korotyaev and Saburova [11], who bound and estimate effective masses at band edges of periodic discrete and metric graphs from geometric parameters, without spanning trees or the Albanese metric. An “effective-mass approximation” of a different kind — the long-distance reduction of the hyperbolic tight-binding Hamiltonian to a Laplace–Beltrami operator — appears in Ref. 2.

For Sec. VI we have found no prior work: we are not aware of any physics literature that places a tight-binding model on an affine building of rank two or on a Ramanujan complex, let alone one that asks whether it has a band structure. A literature search cannot establish a negative, and we state this only as the limit of our knowledge.

II. LATTICES FROM ABELIAN COVERS

Let Y_0 be a finite connected $(q+1)$ -regular graph with n vertices and $m = (q+1)n/2$ edges, E^+ a choice of orientation, and $r = m - n + 1 = b_1(Y_0)$ its cycle rank. Write $\partial: \mathbb{Z}^{E^+} \rightarrow \mathbb{Z}^V$ for the boundary map, $\partial e = t(e) - o(e)$, and $\Delta_0 = \partial\partial^t = (q+1)I - A$ for the graph Laplacian. Let κ_0 denote the number of spanning trees of Y_0 .

Definition 1 (The lattice). A *winding cochain* is a d -tuple $\alpha = (\alpha_1, \dots, \alpha_d)$ with $\alpha_a \in \mathbb{Z}^{E^+}$. The associated lattice $\tilde{Y}_\alpha$ has sites $V \times \mathbb{Z}^d$ and a link from $(o(e), \mathbf{j})$ to $(t(e), \mathbf{j} + \alpha(e))$ for each $e \in E^+$ and each $\mathbf{j} \in \mathbb{Z}^d$. We assume throughout that the period map $H_1(Y_0; \mathbb{Z}) \rightarrow \mathbb{Z}^d$, $c \mapsto (\langle \alpha_a, c \rangle)_a$, is surjective, which is exactly the condition that $\tilde{Y}_\alpha$ be connected; in particular $d \leq r$.

This is not a special construction: *every* $\mathbb{Z}^d$ -periodic tight-binding lattice with n sites per cell and uniform hopping is of this form, with Y_0 the unit cell and $\alpha(e)$ recording how many cells the hopping e crosses. Changing which representative of a site one calls “in cell $\mathbf{0}$ ” replaces α by $\alpha + \partial^t \varphi$; the class $[\alpha] \in H^1(Y_0; \mathbb{Z})^d$ is the gauge-invariant datum.

The Hamiltonian is $H = t\tilde{\Delta}$, with $\tilde{\Delta}$ the Laplacian of $\tilde{Y}_\alpha$: uniform hopping amplitude $-t$ on every link and on-site energy $(q+1)t$. Floquet transform in the $\mathbb{Z}^d$ direction gives $\ell^2(\tilde{Y}_\alpha) \cong \int_{\mathbb{T}^d}^{\oplus} \mathbb{C}^n d^d k / (2\pi)^d$ and $\tilde{\Delta} \cong \int^{\oplus} \Delta_{\mathbf{k}}$ with the Peierls-substituted Bloch Laplacian

$$(\Delta_{\mathbf{k}})_{uv} = (q+1)\delta_{uv} - \sum_{e: u \rightarrow v} e^{i\mathbf{k} \cdot \alpha(e)} - \sum_{e: v \rightarrow u} e^{-i\mathbf{k} \cdot \alpha(e)}, \quad (1)$$

a Hermitian positive-semidefinite $n \times n$ matrix, and bands $E_j(\mathbf{k}) = t\lambda_j(\mathbf{k})$, $\lambda_0 \leq \dots \leq \lambda_{n-1}$. We set $\hbar = 1$ and the lattice constant to unity, so that

$$(M^{-1})_{ab} := \frac{\partial^2 E_0}{\partial k_a \partial k_b} \Big|_{\mathbf{k}=0} = t \frac{\partial^2 \lambda_0}{\partial k_a \partial k_b} \Big|_{\mathbf{k}=0}. \quad (2)$$

Since Δ_0 is a Laplacian of a connected graph its kernel is the constant line, simple, with gap $\lambda_1(0) > 0$; the band bottom sits at $\mathbf{k} = 0$ and λ_0 is analytic near it.

III. THE EFFECTIVE-MASS TENSOR IS THE ALBANESE METRIC

Let $\Pi = I - \partial^t \Delta_0^+ \partial$ be the orthogonal projection of $\mathbb{R}^{E^+}$ onto the cycle space $\ker \partial = H_1(Y_0; \mathbb{R})$, and let $h_a := \Pi \alpha_a$ be the harmonic representative of the a -th winding cochain. Define the *Albanese matrix*

$$H_{ab} := \langle h_a, h_b \rangle = \alpha_a^t \Pi \alpha_b. \quad (3)$$

Theorem 2 (Effective mass). *Under the hypothesis of Definition 1, λ_0 is even in $\mathbf{k}$, simple and analytic near the origin, and*

$$\lambda_0(\mathbf{k}) = \frac{1}{n} \mathbf{k}^t H \mathbf{k} + O(|\mathbf{k}|^4), \quad \text{so} \quad M^{-1} = \frac{2t}{n} H. \quad (4)$$

H is positive definite, and it is the Gram matrix of the classes $[\alpha_a]$ in the flat metric of the Jacobian torus $H^1(Y_0; \mathbb{R})/H^1(Y_0; \mathbb{Z})$, dual to the Albanese torus of Kotani–Sunada [8]: if C is the matrix of a $\mathbb{Z}$ -basis of $H_1(Y_0; \mathbb{Z})$, $G = C^t C$ its Gram matrix and $\mathbf{a}_a = C^t \alpha_a$ the vector of periods, then

$$H_{ab} = \mathbf{a}_a^t G^{-1} \mathbf{a}_b. \quad (5)$$

Equivalently, in terms of the Floquet determinant $\Theta(\mathbf{k}) = \det \Delta_{\mathbf{k}}$,

$$\frac{\partial^2 \Theta}{\partial k_a \partial k_b} \Big|_{\mathbf{k}=0} = 2 \kappa_0 H_{ab}. \quad (6)$$

Proof. $\Delta_{-\mathbf{k}} = \overline{\Delta_{\mathbf{k}}} = \Delta_{\mathbf{k}}^t$, so the spectrum is even in $\mathbf{k}$ and all odd Taylor coefficients of λ_0 vanish. Expanding (1) about $\mathbf{k} = 0$ gives $\Delta_{\mathbf{k}} = \Delta_0 + \sum_a k_a D_a^{(1)} + \frac{1}{2} \sum_{ab} k_a k_b D_{ab}^{(2)} + O(|\mathbf{k}|^3)$ with $(D_a^{(1)})_{uv} = -i \sum_{e: u \rightarrow v} \alpha_a(e) + i \sum_{e: v \rightarrow u} \alpha_a(e)$ and $(D_{ab}^{(2)})_{uv} = \sum_{e: u \rightarrow v} \alpha_a(e) \alpha_b(e) + (u \leftrightarrow v)$. On the constant vector $\mathbf{1}$ one has $D_a^{(1)} \mathbf{1} = i \partial \alpha_a$, $\langle \mathbf{1}, D_a^{(1)} \mathbf{1} \rangle = 0$ and $\langle \mathbf{1}, D_{ab}^{(2)} \mathbf{1} \rangle = 2 \langle \alpha_a, \alpha_b \rangle$. Second-order Rayleigh–Schrödinger perturbation theory at the simple eigenvalue 0 therefore gives

$$\lambda_0(\mathbf{k}) = \frac{1}{n} \sum_{ab} k_a k_b \left(\langle \alpha_a, \alpha_b \rangle - (\partial \alpha_a)^t \Delta_0^+ (\partial \alpha_b) \right) + O(|\mathbf{k}|^4),$$

and the bracket is $\alpha_a^t (I - \partial^t \Delta_0^+ \partial) \alpha_b = H_{ab}$. For (5), h_a is the orthogonal projection of α_a onto the column space of C , so $h_a = C(C^t C)^{-1} C^t \alpha_a$ and $\langle h_a, h_b \rangle = \alpha_a^t C G^{-1} C^t \alpha_b$. Positive definiteness is the surjectivity hypothesis: $\mathbf{v}^t H \mathbf{v} = 0$ forces $\sum_a v_a [\alpha_a] = 0$ in $H^1(Y_0; \mathbb{Q})$, impossible for $\mathbf{v} \neq 0$ when the periods span. Finally $\Theta(\mathbf{k}) = \lambda_0(\mathbf{k}) \prod_{j \geq 1} \lambda_j(\mathbf{k})$, and by the matrix–tree theorem $\prod_{j \geq 1} \lambda_j(0) = n \kappa_0$, which with (4) gives (6). $\square$

For $d = 1$ Eq. (4) is the effective-mass identity of Ref. 19, where $-n \lambda_0''(0)/2$ was identified with an $\mathcal{L}$ -invariant and with a diffusion constant. The content of (5) for $d = 1$ — that the Albanese metric evaluates the harmonic energy — is Kotani–Sunada [7, 9]; what is added here is the $d > 1$ tensor, the identification of the whole object with M^{-1} , and the consequences drawn in Secs. IV–V.

Remark 3. Equation (6) is the practically useful form: $\Theta(\mathbf{k})$ is a Laurent polynomial in $e^{ik_1}, \dots, e^{ik_d}$ with integer coefficients, computable exactly, and its Hessian at the origin delivers $\kappa_0 H$ without any eigenvalue computation. This is how the battery of Sec. VII verifies Theorem 2 in exact arithmetic.

IV. SPANNING TREES AND THE QUANTIZATION OF EFFECTIVE MASS

Theorem 4 (Single-link flux). *Let $d = 1$ and let the winding cochain be the indicator of a single link e , so that the lattice is the unit cell repeated along one direction with exactly one hopping crossing the cell boundary. Then*

$$H = 1 - R_{\text{eff}}(e) = \frac{\tau(Y_0 - e)}{\kappa_0}, \quad (7)$$

where $R_{\text{eff}}(e)$ is the effective resistance across e when every link carries unit resistance, $\kappa_0 = \tau(Y_0)$ is the number of spanning trees of the unit cell, and $\tau(Y_0 - e)$ is the number of spanning trees avoiding e . Consequently

$$M^* = \frac{n}{2t} \cdot \frac{\kappa_0}{\tau(Y_0 - e)}. \quad (8)$$

Proof. $H = \Pi_{ee} = 1 - (\partial^t \Delta_0^+ \partial)_{ee} = 1 - R_{\text{eff}}(e)$ by the standard resistance formula. Kirchhoff’s theorem gives

Y_0	n	m	q	κ_0	$R_{\text{eff}}(e)$	$\tau(Y_0 - e)$	tM^*
K_4	4	6	2	16	1/2	8	4
$K_{3,3}$	6	9	2	81	5/9	36	27/4
Q_3	8	12	2	384	7/12	160	48/5
K_5	5	10	3	125	2/5	75	25/6
Petersen	10	15	2	2000	3/5	800	25/2
Heawood	14	21	2	50421	13/21	19208	147/8

TABLE I. Single-link flux. κ_0 is the number of spanning trees of the unit cell, $\tau(Y_0 - e)$ the number avoiding the flux link, and $tM^* = n \kappa_0 / [2\tau(Y_0 - e)]$ by Eq. (8). For the Heawood graph $\kappa_0 = 3 \cdot 7^5$ and $\tau(Y_0 - e) = 8 \cdot 7^4$. All entries exact.

$\kappa_0 R_{\text{eff}}(e) = \tau(Y_0/e)$, the number of spanning trees containing e , and deletion–contraction $\kappa_0 = \tau(Y_0 - e) + \tau(Y_0/e)$ finishes it. Equation (8) is (4). $\square$

Corollary 5 (Quantization of effective mass). *For any winding cochain, $\kappa_0 H$ is a matrix of integers. Hence every entry of the inverse effective-mass tensor is an integer multiple of $2t/(n \kappa_0)$,*

$$\frac{n \kappa_0}{2t} (M^{-1})_{ab} \in \mathbb{Z}, \quad \det \left[\frac{n \kappa_0}{2t} M^{-1} \right] \in \mathbb{Z}, \quad (9)$$

and the admissible effective masses of a lattice with unit cell Y_0 form a discrete set determined by the single integer κ_0 .

Proof. By Kirchhoff’s theorem in its cycle form, the Gram matrix G of any $\mathbb{Z}$ -basis of $H_1(Y_0; \mathbb{Z})$ has $\det G = \kappa_0$; since $\text{adj } G$ is an integer matrix, $G^{-1} \in \kappa_0^{-1} M_r(\mathbb{Z})$. The period vectors $\mathbf{a}_a = C^t \alpha_a$ are integral, so $\kappa_0 H_{ab} = \mathbf{a}_a^t (\kappa_0 G^{-1}) \mathbf{a}_b \in \mathbb{Z}$ by (5). $\square$

This is the most directly falsifiable statement in the paper. The band curvature at the Γ point of such a lattice cannot be an arbitrary real number: measured in units of $2t/n$, it is a rational with denominator dividing κ_0 . Table I lists the values for six unit cells.

The algebra behind Corollary 5 is elementary, and in the equivalent form “the Albanese lattice of Y_0 has covolume κ_0 ” it is classical [9, 10]. What is at stake here is only what is being quantized: not a lattice covolume but the curvature of a band, which an experiment can measure.

Corollary 6 (Loop flux). *If the winding cochain is the indicator of an oriented cycle $C \subset Y_0$ of length ℓ , then $h = \alpha$, $H = \ell$ exactly, and*

$$M^* = \frac{n}{2t\ell}. \quad (10)$$

The effective mass depends only on the length of the flux loop and on the number of sites per cell — not on the rest of the unit cell.

Proof. A cycle lies in $\ker \partial$, so $\Pi \alpha = \alpha$ and $H = \|\alpha\|^2 = \ell$. $\square$

Y_0	κ_0	$\kappa_0 H$	$\det(\kappa_0 H)$
K_4	16	$\begin{pmatrix} 8 & -4 \\ -4 & 8 \end{pmatrix}$	48
Q_3	384	$\begin{pmatrix} 160 & -32 \\ -32 & 160 \end{pmatrix}$	24576
K_5	125	$\begin{pmatrix} 75 & -25 \\ -25 & 75 \end{pmatrix}$	5000
Heawood	50421	$\begin{pmatrix} 19208 & -9604 \\ -9604 & 19208 \end{pmatrix}$	276710448

TABLE II. The effective-mass tensor for $d = 2$, two single-link fluxes in independent directions. $\kappa_0 H = (n\kappa_0/2t)M^{-1}$ is an integer matrix (Corollary 5); the off-diagonal entries are the mass anisotropy. All entries exact.

Corollary 6 is the sharp opposite of Theorem 4: a single link is maximally sensitive to the ambient cell (through R_{eff}), a closed loop is completely insensitive to it. Both are verified in Sec. VII on K_5 , where the cyclically oriented triangle gives $H = 3$ exactly while the same three links with a non-cyclic orientation give $H = 7/5$.

V. AN EINSTEIN RELATION

The same matrix H governs the classical transport problem on $\tilde{Y}_\alpha$. Let (X_i) be the simple random walk on Y_0 and $\mathbf{S}_N = \sum_{i < N} \boldsymbol{\alpha}(X_i, X_{i+1}) \in \mathbb{Z}^d$ the accumulated winding — the cell coordinate of the lifted walk on $\tilde{Y}_\alpha$.

Theorem 7 (Diffusion tensor). $\mathbf{S}_N/\sqrt{N}$ converges in law to a centred Gaussian with covariance $\sigma_{ab}^2 = H_{ab}/m$, and therefore

$$\sigma^2 = \frac{1}{t(q+1)} M^{-1}. \quad (11)$$

Proof. The walk is reversible with uniform stationary law and transition matrix $A/(q+1)$; the drift is $-\partial\alpha_a/(q+1)$ and the Poisson equation $\Delta_0 g_a = -\partial\alpha_a$ has solution $g_a = -\Delta_0^+ \partial\alpha_a$. Then $\alpha_a + \partial^t g_a = \Pi\alpha_a = h_a$ are the martingale increments, and the martingale central limit theorem gives $\sigma_{ab}^2 = \frac{2}{n(q+1)} \langle h_a, h_b \rangle = H_{ab}/m$. Substituting $H = (n/2t)M^{-1}$ and $m = n(q+1)/2$ gives (11). $\square$

Equation (11) says that the quantum band curvature and the classical diffusion tensor of a lattice of this type differ by the coordination number alone. In particular the mass anisotropy and the diffusion anisotropy are the *same* tensor up to scale, and both are quantized by Corollary 5. The $d = 1$ case is the crystal-lattice central limit theorem of Kotani and Sunada [7], in which the covariance is the Albanese metric; Theorem 2 is what identifies that covariance with a band curvature.

VI. RANK TWO: NO BRILLOUIN ZONE

Everything above needs $b_1(Y_0) \geq 1$. For an ordinary finite graph this is automatic and harmless: a connected

$(q+1)$ -regular graph with $q \geq 2$ has $r = m - n + 1 \geq 2$, so $\mathbb{Z}^d$ covers exist for every $d \leq r$. The hyperbolic lattices of Refs. 1 and 2 are of this type — their unit cells are finite graphs, and the difficulty there is that the *deck* group is non-abelian, not that abelian directions are missing.

We now consider the rank-two analogue. A thick $\tilde{A}_2$ building of order q is the two-dimensional simplicial complex on which PGL_3 of a local field with residue field of size q acts; it is the rank-two counterpart of the $(q+1)$ -regular tree, and every vertex link in it is the incidence graph of the projective plane over $\mathbb{F}_q$ (for $q = 2$, the Heawood graph). Its finite quotients are lattices of sites, links and plaquettes; the Ramanujan complexes of Lubotzky, Samuels and Vishne [14] are among them, and they are built from groups acting simply transitively on the vertices in the sense of Cartwright, Mantero, Steger and Zappa [15]. These are the natural candidates for a higher-rank hyperbolic lattice.

Theorem 8 (No band structure in rank two). *Let $Y = \Gamma \backslash B$ be a finite quotient of a thick $\tilde{A}_2$ building of order $q \geq 2$ by a group Γ acting freely. Then*

- (i) $b_1(Y) = 0$ and $H_1(Y; \mathbb{Z})$ is finite;
- (ii) Y admits no connected $\mathbb{Z}^d$ cover for any $d \geq 1$. There is no $\mathbb{Z}^d$ -periodic lattice with unit cell Y , hence no Brillouin zone, no crystal momentum, and no band structure;
- (iii) the abelian covers of Y are exactly those with deck group a quotient of the finite group $H_1(Y; \mathbb{Z})$, so the threadable flux configurations form the finite group $\text{Hom}(H_1(Y; \mathbb{Z}), U(1)) \cong H_1(Y; \mathbb{Z})$;
- (iv) Γ has Kazhdan's property (T); the trivial representation is isolated in its unitary dual, so no continuous family of unitary characters — abelian or not — deforms the trivial one.

Proof. Γ acts properly and cocompactly on B , so by Garland's vanishing theorem [12] in the form extended to all $q \geq 2$ by Ballmann and Świątkowski [13], $H^1(\Gamma; \mathbb{R}) = 0$ and Γ has property (T). Y is aspherical with $\pi_1(Y) = \Gamma$, so $b_1(Y) = 0$ and $H_1(Y; \mathbb{Z}) = \Gamma^{\text{ab}}$ is finite. A cover with deck group $\mathbb{Z}^d$ requires a surjection $\Gamma \rightarrow \mathbb{Z}^d$, hence a surjection $\Gamma^{\text{ab}} \rightarrow \mathbb{Z}^d$, impossible for $d \geq 1$ from a finite group; this is (ii), and (iii) is the classification of abelian covers by $H^1(Y; \cdot) = \text{Hom}(H_1(Y; \mathbb{Z}), \cdot)$. Part (iv) is property (T). $\square$

Table III makes the statement concrete. On Y_{21} the flux group is $(\mathbb{Z}/2)^6$: there are exactly 64 distinct flux configurations, each link pattern carrying flux 0 or π , and nothing in between. There is no knob to turn continuously, and no momentum to plot a band against.

Remark 9 (What this does *not* say). Theorem 8 is a statement about translational symmetry, not about spectra. The Laplacian and the Hecke operators of Y have

Y	sites	links	plaquettes	b_1	$H_1(Y; \mathbb{Z})$
Y_{21}	21	147	147	0	$(\mathbb{Z}/2)^6$
Y_{24}	24	168	168	0	$(\mathbb{Z}/2)^4 \oplus \mathbb{Z}/14$
Y_{42}	42	294	294	0	$(\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/4)^2$

TABLE III. Three finite quotients of a thick $\tilde{A}_2$ building of order $q = 2$, built as abelian covers of the three-site type quotient of a triangle presentation over $\text{PG}(2, 2)$. Every vertex link is a Heawood graph. $b_1 = 0$ in every case, so none of them is the unit cell of any periodic lattice; the entire flux group is the finite group in the last column. On Y_{21} the only admissible fluxes are 0 and π . All entries exact (p -adic Smith normal forms; see Sec. VII).

perfectly good spectra — indeed the Ramanujan complexes of Ref. 14 have optimal spectral gaps, which is precisely what makes them attractive as synthetic lattices. What fails is the specific device of labelling eigenstates by a crystal momentum in a torus. Any transport theory on a rank-two hyperbolic lattice must be built without a Brillouin zone, and by Theorem 8(iv) not even an infinitesimal one is available near the trivial character. The effective mass of Sec. III is simply not defined there: it is the curvature of a band that does not exist.

Remark 10 (The obstruction is global). The Heawood graph appears on both sides of this paper. As a unit cell in its own right it is unremarkable: $b_1 = 8$, $\kappa_0 = 3 \cdot 7^5$, and a single-link flux gives $tM^* = 147/8$ (Table I). As the *link* of a rank-two complex it is part of a structure with $b_1 = 0$ (Table III). No local inspection of a rank-two hyperbolic lattice reveals the obstruction; it is a property (T) phenomenon, visible only at the level of the fundamental group.

VII. NUMERICAL VERIFICATION

All statements were checked by machine. The script `bloch_mass.py` accompanying this paper runs 90 checks in 13 s and reports every one as passing; its output is archived as `bloch_mass_output.txt`. Exact arithmetic is used throughout except where a row is labelled *numeric*.

A word on the two *numeric* rows. The band identity (4) is checked by evaluating λ_0 at $|\mathbf{k}| = 10^{-6}$ and 5×10^{-7} in 40-digit arithmetic and taking one Richardson step, which removes the $O(|\mathbf{k}|^2)$ truncation and leaves a residual below 10^{-24} ; the observed relative deviations from $\mathbf{k}^t H \mathbf{k} / n$ range from 2×10^{-27} to 2×10^{-26} . The exact row (M) above it, which verifies the Floquet-determinant form (6) by interpolating Θ as a Laurent polynomial with rational coefficients and differentiating it symbolically, is the mathematically complete check; the numeric row is a redundant confirmation that the object being differentiated really is the bottom band.

The rank-two rows use the complex builder and the

key check	#	mode
(S) $H = 1 - R_{\text{eff}}(e)$ and $\kappa_0 H = \tau(Y_0 - e)$, Eq. (7)	6/6	exact
(M) $\partial_{ab}^2 \Theta _0 = 2\kappa_0 H$, Eq. (6)	15/15	exact
(M) $\lambda_0(\mathbf{k}) = \mathbf{k}^t H \mathbf{k} / n + O(\mathbf{k} ^4)$	15/15	numeric
(Q) $\kappa_0 H$ integral, Eq. (9)	15/15	exact
(E) $\sigma^2 = H/m$ by Green-Kubo, Eq. (11)	15/15	exact
(G) gap $\lambda_1(0) > 0$ and $\lambda_0(0) = 0$	15/15	numeric
(X) H reproduces Refs. 16 and 19	4/4	exact
(C) loop flux gives $H = \ell$, Corollary 6	1/1	exact
(B) $b_1 = 0$ and $H_1(Y; \mathbb{Z})$, Table III	4/4	exact
total	90/90	

TABLE IV. The verification battery (`bloch_mass.py`). Unit cells: K_4 , $K_{3,3}$, Q_3 , K_5 , Petersen, Heawood, with single-link fluxes ($d = 1$), two-link fluxes ($d = 2$), and on K_4 and K_5 the multi-chord and triangle patterns of Refs. 16 and 19.

p -adic Smith-normal-form routine of Ref. 17: the complexes of Table III are abelian covers of the three-site type quotient of a triangle presentation over $\text{PG}(2, 2)$ compatible with a Singer cycle, the presentation axioms are re-verified at start-up, and every vertex link is confirmed to be a Heawood graph. The Betti numbers are exact ranks; the torsion subgroups are exact p -adic Smith forms with the prime set certified by the greatest common divisor of the maximal minors.

VIII. DISCUSSION

A. What is testable

Corollary 5 is a constraint on a measurable quantity. For a synthetic lattice assembled from a known unit cell — a coplanar-waveguide network, a photonic-crystal supercell, a cold-atom superlattice — the number κ_0 of spanning trees of the cell is a finite combinatorial datum, and the band curvature at Γ must be an integer multiple of $2t/(n\kappa_0)$. Departures measure either the failure of the uniform-hopping idealization or the presence of longer-range terms; the statement is rigid enough to be diagnostic.

Theorem 4 says more: for a single flux link, the mass is set by the ratio of two spanning-tree counts, so it can be engineered by choosing where the flux link sits, with no computation beyond counting trees. Corollary 6 gives the complementary design rule — route the flux around a closed loop of length ℓ and the mass becomes $n/2t\ell$, insensitive to everything else in the cell.

Equation (11) is a consistency check available without any quantum measurement: the classical diffusion tensor on the same network fixes the effective-mass tensor up to the coordination number.

B. What is open

(a) The non-abelian case. Hyperbolic band theory [2, 3] replaces $\mathbb{T}^d$ by a character variety of a non-abelian surface group. Theorem 2 is a statement about the abelian directions only; whether a “non-abelian effective mass” exists at a higher-genus analogue of the Γ point, and whether it is still a spanning-tree quantity, we do not know.

(b) The quartic term. Band non-parabolicity is the next Taylor coefficient of λ_0 . It is a rational function of the same combinatorial data, but we have no closed form for it and no quantization statement.

(c) Rank two without a Brillouin zone. Theorem 8 removes the tool but not the problem: finite quotients of $\tilde{A}_2$ buildings are excellent expanders with explicit spectra, and the question of what replaces band theory for transport on them — given that property (T) blocks even an infinitesimal character deformation — seems to us the most interesting thing this paper leaves behind.

C. What is impossible

A $\mathbb{Z}^d$ -periodic lattice whose unit cell is a free quotient of a thick $\tilde{A}_2$ building, for any $d \geq 1$ and any $q \geq 2$ (Theorem 8). Any construction claiming a Brillouin zone for such a lattice is in error, and the error will be that the quotient is not free, or the building is not thick, or the complex is not a quotient of a building at all. This is a theorem and does not depend on the state of the literature; the remark in Sec. I, that no one appears to have asked the question, does.

PROVENANCE AND CAVEATS

This is an unrefereed preprint. Every numbered statement is proved from finite linear algebra together

with three classical inputs that are quoted, not improved: Kirchhoff’s matrix–tree theorem in its cycle form, the martingale central limit theorem for additive functionals of a finite reversible chain, and the Garland–Ballmann–Świątkowski vanishing theorem [12, 13]. Kotani–Sunada [7–9] and Baker–Faber [10] supply the Albanese metric, its relation to the spanning-tree count, and the crystal-lattice central limit theorem; Secs. III–V are a physical reading of that material and not a replacement for it.

Attribution, statement by statement. The $d = 1$ case of Theorem 2 is Ref. 19, Theorem 4.2; Eq. (7) for $d = 1$ is Ref. 16, Theorem 1.2(iv); Theorem 8(i) is Ref. 18, Theorem 1.1, which in turn quotes Refs. 12 and 13. The identification of the Albanese metric with a Kirchhoff ratio, and hence the mathematical substance of Corollary 5, is in Refs. 9 and 10. What the present paper adds is the $d > 1$ tensor form, the identification of that tensor with M^{-1} , the deletion–contraction form (8), Corollary 6, the Einstein relation in the form (11), and Sec. VI — the reading of the rank-two vanishing theorem as a no-go theorem for band structure, which is the part we believe to be without precedent. We have not found prior physics work on tight-binding models over rank-two affine buildings; we cannot prove that none exists.

ACKNOWLEDGMENTS

The complex builder and the exact p -adic Smith normal form used in Sec. VII are those of Ref. 17.

-
- [1] A. J. Kollár, M. Fitzpatrick, and A. A. Houck, *Hyperbolic lattices in circuit quantum electrodynamics*, Nature **571**, 45 (2019).
 - [2] J. Maciejko and S. Rayan, *Hyperbolic band theory*, Sci. Adv. **7**, eabe9170 (2021).
 - [3] J. Maciejko and S. Rayan, *Automorphic Bloch theorems for hyperbolic lattices*, Proc. Natl. Acad. Sci. USA **119**, e2116869119 (2022).
 - [4] P. M. Lenggenhager *et al.*, *Simulating hyperbolic space on a circuit board*, Nat. Commun. **13**, 4373 (2022).
 - [5] L. Huang *et al.*, *Hyperbolic photonic topological insulators*, Nat. Commun. **15**, 1647 (2024).
 - [6] X. Xu, A. A. Mahmoud, N. Gorgichuk, R. Thomale, S. Rayan, and M. Mariantoni, *A scalable superconducting circuit framework for emulating physics in hyperbolic space*, arXiv:2510.23827 (2025).
 - [7] M. Kotani and T. Sunada, *Albanese maps and off diagonal long time asymptotics for the heat kernel*, Commun. Math. Phys. **209**, 633 (2000).
 - [8] M. Kotani and T. Sunada, *Standard realizations of crystal lattices via harmonic maps*, Trans. Am. Math. Soc. **353**, 1 (2001).
 - [9] M. Kotani and T. Sunada, *Jacobian tori associated with a finite graph and its abelian covering graphs*, Adv. Appl. Math. **24**, 89 (2000).
 - [10] M. Baker and X. Faber, *Metric properties of the tropical Abel–Jacobi map*, J. Algebr. Comb. **33**, 349 (2011).
 - [11] E. Korotyaev and N. Saburova, *Effective masses for Laplacians on periodic graphs*, J. Math. Anal. Appl. **436**, 104 (2016).
 - [12] H. Garland, *p -adic curvature and the cohomology of discrete subgroups of p -adic groups*, Ann. Math. **97**, 375

- (1973).
- [13] W. Ballmann and J. Świątkowski, *On L^2 -cohomology and property (T) for automorphism groups of polyhedral cell complexes*, Geom. Funct. Anal. **7**, 615 (1997).
 - [14] A. Lubotzky, B. Samuels, and U. Vishne, *Ramanujan complexes of type $\tilde{A}_d$* , Isr. J. Math. **149**, 267 (2005).
 - [15] D. I. Cartwright, A. M. Mantero, T. Steger, and A. Zappa, *Groups acting simply transitively on the vertices of a building of type $\tilde{A}_2$, I*, Geom. Dedicata **47**, 143 (1993).
 - [16] R. Chen, *Algorithmic non-commutative class field theory, IX: exceptional zeros, the graph $\mathcal{L}$ -invariant, and the operator-algebraic Mazur–Tate–Teitelbaum formula*, preprint (2026).
 - [17] R. Chen, *Algorithmic non-commutative class field theory, X: $\tilde{A}_2$ complexes, higher simplicial Laplacians, and the fate of the trace obstruction in rank two*, preprint (2026).
 - [18] R. Chen, *Algorithmic non-commutative class field theory, XII: $\tilde{A}_2$ towers, no abelian sector, the covering norm theorem, and the two digraph towers*, preprint (2026).
 - [19] R. Chen, *Algorithmic non-commutative class field theory, XIV: the graph $\mathcal{L}$ -invariant as spectral action, Bloch effective mass and Albanese length, and what the cyclic Chern character sees*, preprint (2026).